

Theoretical Foundations Syllabus

Math & Theory Track for ML Research

Scope. This is the theoretical-math companion to the main internship prep syllabus. It covers the derivation-level math underlying both the research track (compression, information theory, optimization) and the quant track (probability, statistics, stochastic processes), plus the adjacent theoretical CS material (algorithms, complexity) that both tracks assume. Where a topic is already built out in your comps notes at full derivation depth, it is marked as review rather than repeated in detail — this document instead points to what depth to maintain, and where to go beyond comps scope.

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1 Linear Algebra (Proof-Based)

1.1 Vector spaces and subspaces (*review — already covered in comps notes*)

- Fields, vector space axioms, subspaces

- ☐ Span, linear independence, basis, dimension
- ☐ Change of basis, coordinate representation

1.2 Linear maps and matrices (*review — already covered in comps notes*)

- ☐ Linear transformations as maps, matrix representation
- ☐ Rank, nullity, rank–nullity theorem
- ☐ Invertibility, determinant, and their relationship to rank

1.3 Inner product spaces

- ☐ Inner products, norms induced by inner products, Cauchy–Schwarz
- ☐ Orthogonality, orthogonal complements
- ☐ Gram–Schmidt orthogonalization (derive, don’t just state)
- ☐ Orthogonal projections onto subspaces ties to your least-squares/PCA comps note

1.4 Eigenvalues, eigenvectors, and spectral theory (*review — already covered in comps notes*)

- ☐ Characteristic polynomial, algebraic vs. geometric multiplicity
- ☐ Diagonalizability conditions
- ☐ Spectral theorem for symmetric/Hermitian matrices (know the proof sketch)

1.5 Singular Value Decomposition — full theory

Deepen beyond the comps note: this underlies your low-rank compression experiments.

- ☐ SVD existence proof (via eigendecomposition of $A^T A$)
- ☐ Relationship between SVD and eigendecomposition of symmetric matrices
- ☐ **Eckart–Young theorem:** proof that truncated SVD is the optimal low-rank approximation in Frobenius and spectral norm
- ☐ Pseudo-inverse (Moore–Penrose) via SVD

1.6 Matrix norms, conditioning, and stability

- ☐ Operator norms (spectral norm), Frobenius norm, their relationship to singular values
- ☐ Condition number of a matrix: $\kappa(A) = \sigma_{\max}/\sigma_{\min}$
- ☐ Why ill-conditioning matters for numerical stability (connects to Section 7)

1.7 Matrix calculus

Needed for backprop in matrix/tensor form, not just the scalar case in your comps notes.

- ☐ Gradients of scalar functions w.r.t. vectors and matrices
- ☐ Jacobians of vector-valued functions
- ☐ Chain rule in matrix form (used implicitly every time you backprop through a linear layer)
- ☐ Common identities: $\nabla_x(x^\top Ax)$, $\nabla_A \text{tr}(AB)$, $\nabla_A \log \det A$

1.8 Positive semi-definite matrices

- ☐ Definitions and equivalent characterizations of PSD
- ☐ Cholesky decomposition
- ☐ Why covariance matrices, Gram matrices, and Hessians of convex functions are PSD

2 Real Analysis & Multivariable Calculus

2.1 Single-variable foundations

- ☐ Limits, continuity, differentiability — rigorous ϵ - δ definitions
- ☐ Mean value theorem
- ☐ Sequences and series, convergence tests (ratio, root, comparison)

2.2 Multivariable calculus

- ☐ Partial derivatives, gradients, directional derivatives
- ☐ Hessian matrix, second-order conditions
- ☐ Taylor's theorem — single-variable and multivariable (with remainder term)
- ☐ Chain rule for multivariable functions

2.3 Convexity

This is the theoretical backbone of Section 5 (optimization) — spend real time here.

- ☐ Convex sets: definition, examples, operations that preserve convexity
- ☐ Convex functions: definition via Jensen's inequality
- ☐ First-order condition for convexity ($f(y) \geq f(x) + \nabla f(x)^\top (y - x)$)
- ☐ Second-order condition (Hessian $\succeq 0$)
- ☐ Strict convexity vs. strong convexity, and why strong convexity gives faster optimization rates

2.4 Implicit function theorem (light touch)

- ☐ Statement and intuition
- ☐ Why it matters for constrained optimization (connects to Lagrange multipliers)

3 Probability Theory

3.1 Foundations (*review — already covered in comps notes*)(partially)

- ☐ Probability axioms, sample spaces, events
- ☐ Conditional probability, Bayes' theorem — full fluency, not just the formula
- ☐ Independence vs. conditional independence

3.2 Random variables and distributions

- ☐ Discrete vs. continuous random variables, PMF/PDF/CDF
- ☐ Common discrete distributions: Bernoulli, Binomial, Poisson, Geometric — derive mean/variance for each
- ☐ Common continuous distributions: Uniform, Normal, Exponential, Gamma — derive mean/variance for each
- ☐ Transformations of random variables (change of variables formula)

3.3 Expectation and moments

- ☐ Expectation, variance, linearity of expectation
- ☐ Moment generating functions and characteristic functions
- ☐ Using MGFs to derive moments and prove distributional results (e.g. sums of independent normals)

3.4 Joint distributions

- ☐ Joint, marginal, and conditional distributions
- ☐ Covariance and correlation, covariance matrices
- ☐ Independence in terms of joint density factorization

3.5 Limit theorems

- ☐ Law of Large Numbers (weak and strong — know the distinction)
- ☐ Central Limit Theorem — statement, intuition, and where it breaks down

3.6 Concentration inequalities

High-value for both ML theory (generalization bounds) and quant interviews.

- ☐ Markov's inequality
- ☐ Chebyshev's inequality (derive from Markov)
- ☐ Hoeffding's inequality
- ☐ Chernoff bounds — derivation via MGF and optimization over a free parameter

4 Mathematical Statistics

4.1 Point estimation

- ☐ Maximum likelihood estimation (*review — already covered in comps notes*) extend beyond the logistic-regression case in comps to the general framework
- ☐ Method of moments
- ☐ Sufficiency and the factorization theorem

4.2 Estimator quality

- ☐ Bias, variance, mean squared error decomposition (*review — already covered in comps notes*)
- ☐ Consistency and asymptotic efficiency
- ☐ Cramér–Rao lower bound (Fisher information)

4.3 Hypothesis testing theory

- ☐ Null/alternative hypotheses, Type I/II errors, power (*review — already covered in comps notes*)
- ☐ Neyman–Pearson lemma (most powerful tests)
- ☐ Likelihood ratio tests
- ☐ p -values — correct interpretation (and common misinterpretations to avoid in interviews)

4.4 Interval estimation

- ☐ Confidence intervals — construction and interpretation
- ☐ Relationship between confidence intervals and hypothesis tests

4.5 Bayesian statistics

- ☐ Prior, likelihood, posterior — Bayes' rule for parameters
- ☐ Conjugate priors (Beta–Binomial, Normal–Normal)
- ☐ MAP estimation vs. MLE — when they coincide

5 Optimization Theory

5.1 Unconstrained optimization

- ☐ First-order and second-order optimality conditions
- ☐ Gradient descent: convergence rate for convex and strongly convex functions (derive the rates, don't just cite them)
- ☐ Effect of the Lipschitz constant and step size choice on convergence

5.2 Constrained optimization

- ☐ Lagrange multipliers for equality constraints
- ☐ Karush–Kuhn–Tucker (KKT) conditions for inequality constraints — derive, don't memorize
- ☐ Complementary slackness

5.3 Duality

- ☐ Lagrangian dual function and dual problem
- ☐ Weak duality (always holds) vs. strong duality (needs conditions)
- ☐ Slater's condition
- ☐ Revisit SVM duality (from comps) as a worked example of all of the above

5.4 Convergence theory for iterative methods

- ☐ Convergence rate of momentum methods
- ☐ Nesterov acceleration — what changes vs. vanilla momentum
- ☐ SGD convergence under noisy gradients — bias/variance of the stochastic gradient estimator

5.5 Second-order methods (conceptual)

- ☐ Newton's method — quadratic convergence, cost of the Hessian
- ☐ Quasi-Newton methods (BFGS) — how they approximate the Hessian cheaply

6 Information Theory

This is the theoretical backbone of *why* compression works at all — directly relevant to your research project.

6.1 Entropy

- ☐ Shannon entropy — definition and interpretation as expected information content
- ☐ Joint entropy, conditional entropy, chain rule for entropy

6.2 Divergence and cross-entropy

- ☐ Cross-entropy — relationship to entropy and KL divergence
- ☐ KL divergence — definition, non-negativity (Gibbs' inequality), and why it's asymmetric
- ☐ Why KL divergence is the natural loss for distillation (connects to Track C/D in the main syllabus)

6.3 Mutual information

- ☐ Definition in terms of KL divergence between joint and product-of-marginals
- ☐ Relationship to entropy: $I(X; Y) = H(X) - H(X|Y)$

6.4 Rate-distortion theory (conceptual)

- ☐ The rate-distortion tradeoff — formal statement of the accuracy/compression tradeoff you're studying empirically
- ☐ Why this gives a theoretical lower bound to compare your empirical Pareto frontier against

7 Numerical Methods & Numerical Linear Algebra

7.1 Floating point and numerical error

- ☐ Floating point representation (FP32/FP16/BF16) — mantissa/exponent tradeoffs
- ☐ Rounding error, catastrophic cancellation
- ☐ Why BF16 preserves training stability better than FP16 (exponent range vs. mantissa precision)

7.2 Conditioning and stability

- ☐ Condition number and error amplification
- ☐ Numerically stable vs. unstable algorithms for the same problem (e.g. computing variance)

7.3 Iterative numerical linear algebra

- ☐ Power iteration for dominant eigenvector/eigenvalue
- ☐ Conceptual overview of Lanczos / randomized SVD for large-scale low-rank approximation

7.4 Quantization as numerical approximation

- ☐ Uniform vs. non-uniform quantization — formalize as a rounding operator
- ☐ Quantization error as additive noise — bias and variance of the quantization error model
- ☐ Effect of quantization noise on gradient flow (why QAT needs a straight-through estimator)

8 Stochastic Processes (Quant Track)

8.1 Markov chains

- ☐ Transition matrices, Chapman–Kolmogorov equations
- ☐ Stationary distributions, existence and uniqueness conditions
- ☐ Ergodicity, mixing

8.2 Random walks

- ☐ Simple random walk — expected position, variance growth
- ☐ Gambler’s ruin as a worked example (classic quant interview problem)

8.3 Martingales (conceptual)

- ☐ Definition and intuition (fair-game property)
- ☐ Stopping times, optional stopping theorem — statement and intuition

8.4 Brownian motion and Itô calculus (conceptual)

- ☐ Brownian motion — defining properties (independent increments, continuity)
- ☐ Itô’s lemma — statement and why it differs from the ordinary chain rule
- ☐ Enough intuition to reason about it in an interview, not full stochastic calculus fluency

9 Discrete Math & Combinatorics

9.1 Counting

- ☐ Permutations, combinations, binomial coefficients
- ☐ Inclusion–exclusion principle
- ☐ Pigeonhole principle

9.2 Recurrences and generating functions (light)

- ☐ Solving linear recurrences
- ☐ Generating functions as a counting tool (conceptual)

9.3 Graph theory (theoretical)

- ☐ Trees, connectivity, cycles
- ☐ Graph coloring — basic bounds

10 Theory of Algorithms & Complexity

10.1 Asymptotic analysis

- Big-O, Ω , Θ — formal definitions, not just intuition
- Amortized analysis (e.g. dynamic array resizing)

10.2 Complexity classes (conceptual)

- P vs. NP — definitions and why the distinction matters practically
- NP-completeness — what a reduction proof looks like, at a high level

10.3 Algorithmic paradigms with correctness proofs

- Divide and conquer — recurrence-based analysis (Master theorem)
- Dynamic programming — optimal substructure and overlapping subproblems, formal justification
- Greedy algorithms — exchange argument proofs of optimality

10.4 Graph algorithms theory

- Shortest path algorithms — correctness proof sketch for Dijkstra
- Minimum spanning tree — correctness of Kruskal's/Prim's (cut property)

Suggested Sequencing

Work top-to-bottom roughly in this order, since later sections lean on earlier ones:

1. Linear Algebra (Section 1) $\rightarrow$ Real Analysis & Convexity (Section 2) $\rightarrow$ Optimization Theory (Section 5) — these three form one continuous arc.
2. Probability (Section 3) $\rightarrow$ Mathematical Statistics (Section 4) — same arc, run in parallel with the above.
3. Information Theory (Section 6) once Sections 1 and 3 are solid — it draws on both.
4. Numerical Methods (Section 7) alongside your compression experiments, as topics become practically relevant.
5. Stochastic Processes (Section 8), Discrete Math (Section 9), and Algorithms/Complexity (Section 10) run as the dedicated quant-interview-prep block, in parallel with everything above rather than after it.